A Categorized Reference of Statistical Tests & Methods

From t-tests, ANOVA, and GLMs to effect sizes, power, causal inference, and beyond

This document groups commonly used statistical tests and modeling procedures by purpose. It is intended as a practical map of the field rather than a truly exhaustive catalogue — statistics is an active discipline and new and specialized procedures are introduced regularly. A handful of entries (e.g., AIC/BIC, VIF, Cronbach's alpha, I^2) are measures or criteria rather than formal hypothesis tests, but are included because they are routinely reported alongside the tests in their family.

Choosing a test depends on the research question, the measurement scale of the variables, the number and independence of groups, and whether distributional assumptions are met. When parametric assumptions fail, consult the non-parametric or resampling alternatives in Sections 2 and 18. Cross-cutting foundations that apply to nearly every test — the kinds of decision error (Type I, II, III, S, M), effect-size measures, and statistical power — are covered in Sections 21-23, with confidence intervals, diagnostic accuracy, missing data, causal inference, and several specialized domains following them.

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1. Tests of central tendency (parametric, means)

Compare means of one or more groups assuming approximate normality and (often) equal variances.

Test	Primary purpose	Notes / key assumptions
One-sample t-test	Compare a sample mean to a known/hypothesized value	Normality of the mean; small n sensitive to outliers
Independent two-sample t-test	Compare means of two independent groups	Assumes equal variances; normality within groups
Welch's t-test	Two-group mean comparison without equal-variance assumption	Preferred default when variances may differ
Paired-samples t-test	Compare means of two related/matched measurements	Tests mean of paired differences; differences ~ normal
One-sample z-test	Mean vs. value when population SD is known / large n	Rarely used in practice; needs known σ
Two-sample z-test	Compare two means with known σ or large samples	Proportion z-test is a special case
One-way ANOVA	Compare means across 3+ independent groups	Equal variances, normal residuals, independence
Welch's ANOVA	3+ group means with unequal variances	Robust alternative to one-way ANOVA
Factorial / two-way ANOVA	Effects of two+ factors and their interactions	Tests main effects and interaction terms
Repeated-measures ANOVA	Within-subject means across conditions/time	Assumes sphericity (see Mauchly's test)
Mixed-design ANOVA	Between- and within-subject factors together	Combines independent and repeated factors
ANCOVA	Group means adjusting for a continuous covariate	Assumes homogeneity of regression slopes
MANOVA	Compare groups on several dependent variables jointly	Uses Wilks' Λ, Pillai, Hotelling-Lawley, Roy
MANCOVA	MANOVA with covariate adjustment	Multivariate extension of ANCOVA
Hotelling's T^2	Two-group comparison of a mean vector	Multivariate analogue of the two-sample t-test

2. Non-parametric tests (distribution-free)

Rank- or sign-based alternatives used when normality fails or data are ordinal.

Test	Primary purpose	Notes / key assumptions
Mann-Whitney U (Wilcoxon rank-sum)	Compare two independent groups on ranks	Tests stochastic dominance / distribution shift
Brunner-Munzel test	Compare two groups without assuming equal distribution shapes	Generalizes Mann-Whitney; tests $P(X<Y) \neq ½$
Wilcoxon signed-rank test	Compare two related samples (paired) on ranks	Non-parametric paired t-test alternative
Sign test	Paired comparison using only direction of differences	Weak power; minimal assumptions

Test	Primary purpose	Notes / key assumptions
Kruskal-Wallis test	Compare 3+ independent groups on ranks	Non-parametric one-way ANOVA
Friedman test	Compare 3+ related/repeated conditions on ranks	Non-parametric repeated-measures ANOVA
Mood's median test	Compare medians across groups	Less powerful but very robust
Jonckheere-Terpstra test	Ordered/trend differences across groups	For a priori ordered alternatives
Page's trend test	Ordered trend across related conditions	Repeated-measures trend analogue
Cochran's Q test	Compare 3+ matched binary outcomes	Extension of McNemar for k conditions
Wald-Wolfowitz runs test	Randomness / difference of two distributions	Based on runs in pooled ordering

3. Categorical & frequency / proportion tests

Analyze counts, proportions, and contingency tables.

Test	Primary purpose	Notes / key assumptions
Chi-squared test of independence	Association between two categorical variables	Expected counts ≥ 5; large-sample approximation
Chi-squared goodness-of-fit	Observed frequencies vs. expected distribution	Compares categorical distribution to theory
Fisher's exact test	Association in small 2×2 (or r×c) tables	Exact; preferred for small expected counts
Barnard's test	2×2 association, unconditional exact test	Often more powerful than Fisher for 2×2
Boschloo's test	Exact 2×2 association	Uniformly more powerful than Fisher's exact test
G-test (likelihood-ratio chi-squared)	Goodness-of-fit / independence	Uses likelihood ratio instead of Pearson statistic
McNemar's test	Change in paired binary outcomes	Marginal homogeneity in 2×2 paired data
Cochran-Mantel-Haenszel	Association across stratified 2×2 tables	Controls for a confounding stratum
Binomial test	Single proportion vs. hypothesized value	Exact test for one binary variable
One/two-proportion z-test	Compare proportions to a value or each other	Large-sample normal approximation
Multinomial test	Counts across >2 categories vs. expected	Exact analogue of goodness-of-fit
Cochran-Armitage trend test	Trend in proportions across ordered groups	Dose-response in 2×k tables

4. Correlation & measures of association

Quantify and test the strength/direction of relationships.

Test	Primary purpose	Notes / key assumptions
Pearson correlation (r)	Linear association between two continuous variables	Sensitive to outliers; assumes bivariate normality for inference
Spearman's rank correlation (ρ)	Monotonic association on ranks	Non-parametric; robust to outliers
Kendall's tau	Ordinal association via concordant pairs	Good for small samples / many ties
Point-biserial correlation	Continuous vs. dichotomous variable	Special case of Pearson r
Partial correlation	Association controlling for other variables	Removes linear effect of covariates
Phi coefficient	Association in a 2×2 table	Equivalent to Pearson on binary data
Cramér's V	Association strength in r×c tables	Normalizes chi-squared to 0-1
Goodman-Kruskal gamma	Ordinal association	Ignores ties between pairs
Distance correlation	Detects any (incl. nonlinear) dependence	Zero iff variables are independent
Intraclass correlation (ICC)	Agreement/consistency of repeated measures	Several forms (ICC(1), (2,k), etc.)

5. Regression & the generalized linear model (GLM) family

Model an outcome as a function of predictors; the GLM unifies many of these via a link function and error distribution.

Test	Primary purpose	Notes / key assumptions
Simple / multiple linear regression (OLS)	Continuous outcome from one+ predictors	Linearity, independence, homoscedasticity, normal errors
Logistic regression	Binary outcome probability	Logit link, binomial family
Multinomial logistic regression	Nominal outcome with 3+ categories	Generalizes logistic to unordered classes
Ordinal (proportional-odds) regression	Ordered categorical outcome	Assumes proportional odds
Probit regression	Binary outcome via normal CDF link	Alternative to logit
Poisson regression	Count outcomes / rates	Assumes mean = variance
Negative binomial regression	Overdispersed count outcomes	Adds dispersion parameter vs. Poisson
Quasi-Poisson regression	Overdispersed counts via scaling	Adjusts standard errors, not full distribution
Zero-inflated (ZIP / ZINB) models	Counts with excess zeros	Mixture of a count and a zero-generating process
Gamma / inverse-Gaussian regression	Positive, right-skewed continuous outcomes	GLM with appropriate link
Tobit regression	Censored continuous outcomes	Handles floor/ceiling censoring
Ridge / Lasso / Elastic-net	Regression with regularization	Shrinkage for collinearity / variable selection
Quantile regression	Model conditional quantiles (e.g., median)	Robust; no normality needed

Test	Primary purpose	Notes / key assumptions
GLMM (mixed-effects GLM)	Clustered/hierarchical data with random effects	Handles repeated measures, nesting
GEE	Population-averaged models for correlated data	Specifies a working correlation structure
Nonlinear regression	Outcome via a nonlinear functional form	Requires starting values; iterative fit

6. Model comparison & significance of parameters

General-purpose tests for comparing nested models or testing coefficients.

Test	Primary purpose	Notes / key assumptions
Likelihood-ratio test (LRT)	Compare nested models via log-likelihoods	Asymptotic chi-squared distribution
Wald test	Test individual coefficients / restrictions	Uses estimate and its standard error
Score / Lagrange-multiplier test	Test restrictions using the score function	Only needs the restricted model
F-test (nested models)	Compare nested linear models	Extra-sum-of-squares principle
Vuong test	Compare non-nested models	Used e.g. for zero-inflated vs. standard counts
AIC / BIC	Information criteria for model selection	Not hypothesis tests; penalize complexity
Omnibus / overall F-test	Whether a model explains variance overall	Tests all slopes jointly = 0

7. Normality & distributional goodness-of-fit

Assess whether data follow a specified (often normal) distribution.

Test	Primary purpose	Notes / key assumptions
Shapiro-Wilk test	Normality, especially small-to-moderate n	Powerful general normality test
Shapiro-Francia test	Normality (simplified Shapiro-Wilk)	Order-statistic based; suited to larger n
Kolmogorov-Smirnov test	Fit to any continuous distribution	One- and two-sample versions; weak in tails
Lilliefors test	Normality with estimated parameters	K-S corrected for estimated mean/SD
Anderson-Darling test	Distributional fit with tail emphasis	More sensitive in the tails than K-S
Cramér-von Mises test	Distributional goodness-of-fit	Integrated squared CDF difference
Jarque-Bera test	Normality via skewness and kurtosis	Large-sample; common in econometrics
D'Agostino's K^2 test	Normality via skewness + kurtosis	Omnibus moment-based test
Ryan-Joiner test	Normality via correlation with normal scores	Similar to Shapiro-Wilk
Chi-squared goodness-of-fit	Binned fit to a distribution	Needs adequate expected bin counts

Test	Primary purpose	Notes / key assumptions
Mardia's test	Multivariate normality via skewness & kurtosis	Standard check before MANOVA / discriminant analysis
Henze-Zirkler test	Multivariate normality (consistent omnibus)	Based on a weighted distance of the empirical CF
Royston's test	Multivariate normality	Extends Shapiro-Wilk to several variables

8. Homogeneity of variance / scale

Check the equal-variance assumption used by many parametric tests, or test directly for differences in spread (including rank-based, non-parametric scale tests).

Test	Primary purpose	Notes / key assumptions
F-test for two variances	Compare two group variances	Very sensitive to non-normality
Levene's test	Equality of variances across groups	Robust; uses deviations from group mean
Brown-Forsythe test	Equality of variances (median-based)	More robust variant of Levene
Bartlett's test	Equality of variances across groups	Powerful but assumes normality
Fligner-Killeen test	Equality of variances, rank-based	Highly robust to departures from normality
Siegel-Tukey test	Difference in scale/spread between two groups	Rank-based; non-parametric
Ansari-Bradley test	Difference in dispersion between two groups	Rank-based; assumes equal medians
Mood's test for scale	Difference in dispersion between two groups	Rank-based scale test (distinct from Mood's median test)
Hartley's F-max test	Ratio of largest to smallest variance	Requires equal group sizes; sensitive
Mauchly's test of sphericity	Sphericity in repeated-measures ANOVA	Violations corrected via Greenhouse-Geisser
Box's M test	Equality of covariance matrices	Used before MANOVA/discriminant analysis

9. Data transformations & variance stabilization

These reshape a variable's distribution — to stabilize variance, reduce skew, linearize relationships, or meet the normality assumptions of the tests above. Like the methods in later sections, they are procedures applied to data rather than hypothesis tests.

Test	Primary purpose	Notes / key assumptions
Log transformation (ln, log10)	Reduce right skew; stabilize variance	Strictly positive values; turns multiplicative into additive
Square-root transformation	Stabilize variance for count data	Mild; for Poisson-like counts / moderate skew
Reciprocal (1/x) transformation	Strong skew reduction; compress large values	Positive values; reverses rank order

Test	Primary purpose	Notes / key assumptions
Box-Cox transformation	Estimate the optimal power transform	Positive data only; λ chosen by maximum likelihood
Yeo-Johnson transformation	Power transform allowing zero/negative values	Generalizes Box-Cox to the whole real line
Inverse hyperbolic sine (IHS / arcsinh)	Log-like transform defined at 0 and negatives	Common in economics for skewed income/wealth with zeros
Arcsine (arcsine-root) transformation	Stabilize variance of proportions/percentages	For [0,1] data; now often discouraged vs. GLMs
Logit transformation	Map proportions in (0,1) to the real line	log(p/(1-p)); underlies logistic regression
Tukey's ladder of powers	Systematic search across power transforms	Re-expression to guide transform choice
Fisher z-transformation	Stabilize variance of a correlation coefficient	Enables confidence intervals / tests on Pearson r
Rank-based inverse-normal (rankit)	Force a variable toward normality via ranks	Nonparametric normalization; common in GWAS/genetics
Anscombe / Freeman-Tukey transform	Variance stabilization for count data	Refinements of the square-root transform
Standardization (z-score)	Center and scale to mean 0, SD 1	Aids comparability; does not correct skew
Min-max normalization	Rescale to a fixed range (e.g., 0-1)	Sensitive to outliers; common in ML preprocessing
Differencing	Remove trend/seasonality for stationarity	Time-series preprocessing (see Section 12)

10. Post-hoc & multiple-comparison procedures

Follow-up comparisons after an omnibus test. Most procedures here control the family-wise error rate (FWER) — the probability of at least one false positive across the family of tests — while the Benjamini-Hochberg procedure instead controls the less stringent false discovery rate (FDR), the expected proportion of false positives among rejections.

Test	Primary purpose	Notes / key assumptions
Tukey HSD	All pairwise mean comparisons after ANOVA	Controls FWER; equal n ideal
Bonferroni correction	Adjust α across many comparisons (FWER)	Conservative; widely applicable
Holm-Bonferroni	Sequentially rejective FWER control	Uniformly more powerful than Bonferroni
Hochberg's step-up procedure	Step-up FWER control	More powerful than Holm under independence/positive dependence
Hommel's procedure	FWER control via closed testing	More powerful than Hochberg; more computation
Šidák correction	FWER α adjustment assuming independence	Slightly less conservative than Bonferroni

Test	Primary purpose	Notes / key assumptions
Scheffé's method	Arbitrary contrasts after ANOVA	Very conservative; flexible contrasts
Dunnett's test	Compare treatments to a single control	Controls error for many-vs-control
Games-Howell	Pairwise comparisons with unequal variances	Does not assume equal variances/n
Newman-Keuls / Duncan's	Stepwise pairwise comparisons	More powerful but less strict error control
Dunn's test	Pairwise after Kruskal-Wallis	Non-parametric multiple comparison
Nemenyi test	Pairwise after Friedman test	Non-parametric, rank-based
Benjamini-Hochberg (FDR)	Control the false discovery rate	Less stringent than FWER; standard for high-dimensional testing
Benjamini-Yekutieli (FDR)	FDR control under arbitrary dependence	More conservative variant of Benjamini-Hochberg

11. Regression diagnostics & specification tests

Validate assumptions of fitted regression models.

Test	Primary purpose	Notes / key assumptions
Breusch-Pagan test	Heteroscedasticity of residuals	Regresses squared residuals on predictors
White's test	Heteroscedasticity (general form)	No specific functional form assumed
Goldfeld-Quandt test	Heteroscedasticity across a split	Compares variance in two subsamples
Durbin-Watson test	First-order residual autocorrelation	Statistic near 2 indicates no autocorrelation
Breusch-Godfrey test	Higher-order serial correlation	LM test; more general than Durbin-Watson
Ramsey RESET test	Functional-form misspecification	Detects omitted nonlinearity
Variance inflation factor (VIF)	Multicollinearity among predictors	Diagnostic measure, not a formal test
Chow test	Structural break / parameter stability	Tests equality of coefficients across groups
Hausman test	Fixed vs. random effects / endogeneity	Compares consistent and efficient estimators
Harvey-Collier test	Linearity of the model	Tests recursive residuals
Hosmer-Lemeshow test	Goodness-of-fit / calibration for logistic regression	Groups by predicted-risk deciles; criticized for low power
Cook's distance	Influential observations	Per-point influence diagnostic

12. Time-series tests

Stationarity, autocorrelation, cointegration, and causality in temporal data.

Test	Primary purpose	Notes / key assumptions
Augmented Dickey-Fuller (ADF)	Unit root / non-stationarity	Null = unit root present
Phillips-Perron test	Unit root, robust to serial correlation	Non-parametric correction to ADF
KPSS test	Stationarity (null = stationary)	Complements ADF; reversed hypothesis
Ljung-Box test	Autocorrelation at multiple lags	Portmanteau test on residuals
Box-Pierce test	Autocorrelation (portmanteau)	Predecessor to Ljung-Box
Granger causality test	Whether one series helps predict another	Predictive, not true causality
Engle's ARCH test	Autoregressive conditional heteroscedasticity	Volatility clustering in residuals
Johansen test	Cointegration among multiple series	Determines number of cointegrating vectors
Engle-Granger test	Cointegration of two series	Two-step residual-based approach
Mann-Kendall trend test	Monotonic trend in a time series	Non-parametric; common in environmental data
Cox-Stuart test	Trend via sign comparisons	Simple distribution-free trend test
Diebold-Mariano test	Compare the accuracy of two forecasts	Tests equal expected loss; not for nested models
Variance-ratio test (Lo-MacKinlay)	Whether a series follows a random walk	Common in finance / market-efficiency studies
CUSUM / CUSUM-of-squares test	Parameter stability over time	Detects gradual structural change in residuals
Bai-Perron test	Multiple structural breaks at unknown dates	Estimates number and timing of breaks
Zivot-Andrews test	Unit root allowing one structural break	Endogenously determines the break date

13. Survival / time-to-event analysis

Handle censored time-to-event data.

Test	Primary purpose	Notes / key assumptions
Kaplan-Meier estimator	Estimate survival function	Non-parametric; handles right-censoring
Log-rank test	Compare survival curves across groups	Equal weight to all time points
Gehan-Breslow (Wilcoxon)	Compare survival curves	Weights early events more heavily
Tarone-Ware test	Compare survival curves	Intermediate weighting scheme
Cox proportional-hazards model	Effect of covariates on hazard	Assumes proportional hazards
Schoenfeld residuals test	Proportional-hazards assumption check	Tests time-dependence of effects

Test	Primary purpose	Notes / key assumptions
Accelerated failure time (AFT)	Parametric survival modeling	Models effect on event time directly
Competing-risks analysis (Fine-Gray)	Subdistribution hazards with competing events	Models cumulative incidence functions
Frailty models	Survival with random effects / clustering	Account for unobserved heterogeneity
Extended Cox (time-varying covariates)	Covariates that change over follow-up	Relaxes proportional hazards for such covariates
Restricted mean survival time (RMST)	Average event-free time up to a horizon	Useful alternative when hazards are non-proportional

14. Multivariate hypothesis tests (mean vectors & covariance)

Tests on mean vectors, covariance structure, and group separation.

Test	Primary purpose	Notes / key assumptions
Wilks' lambda	Overall MANOVA significance	Likelihood-ratio statistic
Pillai's trace	MANOVA significance (most robust)	Preferred under assumption violations
Hotelling-Lawley trace / Roy's largest root	MANOVA significance	Alternative multivariate criteria
Bartlett's test of sphericity	Whether correlation matrix ≠ identity	Prerequisite for factor analysis/PCA
Kaiser-Meyer-Olkin (KMO)	Sampling adequacy for factor analysis	Measure, not a formal hypothesis test
PERMANOVA	Group differences in multivariate distance	Permutation-based; common in ecology
Mantel test	Correlation between two distance matrices	Permutation-based
Mahalanobis distance	Multivariate outlier / distance	Accounts for covariance structure

15. Dimension reduction & latent-structure methods

These are multivariate analysis methods and matrix decompositions rather than hypothesis tests — they summarize structure, extract components, or group cases. They are included because they sit alongside the multivariate tests above and are frequently part of the same workflow.

Test	Primary purpose	Notes / key assumptions
Principal component analysis (PCA)	Reduce dimensionality via variance-maximizing orthogonal components	Unsupervised; components ranked by explained variance
Independent component analysis (ICA)	Separate a signal into statistically independent sources	Assumes non-Gaussian independent components (blind source separation)
Exploratory factor analysis (EFA)	Recover latent factors behind observed correlations	Check Bartlett's sphericity & KMO first (Section 14)

Test	Primary purpose	Notes / key assumptions
Confirmatory factor analysis (CFA)	Test a hypothesized latent-factor structure	Estimated within structural equation modeling
Structural equation modeling (SEM)	Model relationships among latent and observed variables	Combines measurement and path models
Linear discriminant analysis (LDA)	Find axes that best separate known groups	Supervised; assumes equal covariance matrices
Canonical correlation analysis (CCA)	Relationships between two sets of variables	Finds maximally correlated linear combinations
Partial least squares (PLS)	Dimension reduction targeted at predicting an outcome	Useful when predictors outnumber observations
Multidimensional scaling (MDS)	Low-dimensional layout preserving distances	Input is a distance / dissimilarity matrix
Correspondence analysis	Reduce dimensionality of categorical/contingency data	PCA analogue for frequency tables
t-SNE / UMAP	Nonlinear embedding for visualization	Manifold-learning methods; mainly exploratory
Cluster analysis (k-means, hierarchical)	Group observations by similarity	Unsupervised; number of clusters often must be chosen

16. Agreement, reliability & equivalence

Quantify rater agreement, internal consistency, and statistical equivalence.

Test	Primary purpose	Notes / key assumptions
Cohen's kappa	Agreement between two raters (categorical)	Corrects for chance agreement
Weighted kappa	Ordinal agreement between two raters	Penalizes disagreements by distance
Fleiss' kappa	Agreement among 3+ raters	Extension of Cohen's kappa
Krippendorff's alpha	Agreement across raters/data types	Handles missing data and any scale
Cronbach's alpha	Internal consistency of a scale	Reliability coefficient, not a test
Bland-Altman analysis	Agreement between two measurement methods	Plots difference vs. mean
TOST (two one-sided tests)	Statistical equivalence within bounds	Tests for absence of meaningful difference
Kendall's W (coefficient of concordance)	Agreement among raters on rankings	0 = no agreement, 1 = complete agreement

17. Outlier & influence tests

Detect anomalous observations.

Test	Primary purpose	Notes / key assumptions
Grubbs' test	Single outlier in a normal sample	Iterative for multiple outliers

Test	Primary purpose	Notes / key assumptions
Dixon's Q test	Outlier in a small sample	Simple ratio-based test
Rosner's ESD test	Multiple outliers (generalized ESD)	Specify maximum number of outliers
Tietjen-Moore test	Known number of outliers	Requires specifying count in advance
Cook's distance / leverage	Influential points in regression	Diagnostic measures

18. Resampling, exact & Bayesian approaches

Computationally intensive or probability-based alternatives to classical tests.

Test	Primary purpose	Notes / key assumptions
Bootstrap	Estimate sampling distribution by resampling	Few distributional assumptions
Permutation / randomization test	Exact p-values by reshuffling labels	Valid under exchangeability
Jackknife	Bias and variance estimation	Leave-one-out resampling
Monte Carlo test	Approximate exact tests via simulation	Used when analytic null is intractable
Bayes factor	Evidence ratio between two hypotheses	Bayesian alternative to p-values
Bayesian estimation (e.g., BEST)	Posterior comparison of groups	Credible intervals instead of p-values
Posterior predictive checks	Model fit assessment in Bayesian models	Compares simulated vs. observed data
MCMC (Metropolis-Hastings, Gibbs, HMC/NUTS)	Sample from a posterior distribution	Engine behind most modern Bayesian inference
Gelman-Rubin diagnostic (R-hat)	Assess MCMC convergence across chains	Values near 1.0 indicate convergence
Effective sample size (ESS)	Information content of correlated MCMC draws	Low ESS signals poor mixing
WAIC / LOO-CV	Bayesian out-of-sample predictive accuracy	Modern alternatives to DIC
Deviance information criterion (DIC)	Bayesian model comparison	Penalizes effective number of parameters
Credible / highest-density (HPD) interval	Bayesian interval estimate	Direct probability statement about a parameter

19. Meta-analysis & heterogeneity tests

Combine and assess consistency across studies.

Test	Primary purpose	Notes / key assumptions
Cochran's Q (heterogeneity)	Whether study effects differ	Low power with few studies

Test	Primary purpose	Notes / key assumptions
I^2 statistic	Proportion of variation due to heterogeneity	Descriptive index, not a test
Egger's regression test	Funnel-plot asymmetry / publication bias	Regression-based
Begg's rank test	Publication bias via rank correlation	Non-parametric alternative to Egger

20. Error types & statistical decision errors

Foundational concepts describing the ways a statistical decision can be wrong. These underpin significance testing, power, and multiple-comparison corrections rather than being tests themselves.

Error / concept	What it is	Practical implication
Type I error (α)	Rejecting a true null hypothesis (false positive)	Set by the significance level; inflated by multiple testing
Type II error (β)	Failing to reject a false null (false negative)	Reduced by larger samples / effects; $1-\beta$ is power
Type III error	Correctly rejecting the null but inferring the wrong direction	Right conclusion of 'an effect exists', wrong sign
Type IV error	Misinterpreting a correctly rejected null	Classic example: misreading an interaction effect
Type S error (sign)	Estimating the wrong sign of an effect	Gelman-Carlin framework; likely under low power
Type M error (magnitude)	Exaggerating the size of an effect	Significant low-power studies overstate effect sizes
Family-wise error rate (FWER)	Probability of ≥1 Type I error across a family	Controlled by Bonferroni/Holm etc. (Section 10)
False discovery rate (FDR)	Expected share of false positives among rejections	Controlled by Benjamini-Hochberg (Section 10)
p-value	Probability of data as extreme as observed, if null true	Not the probability the null is true; threshold-dependent

21. Effect-size measures

Quantify the magnitude of an effect, independent of sample size. A p-value indicates whether an effect exists; the effect size indicates whether it matters, and each test family has a natural companion measure.

Measure	What it quantifies	Typical context
Cohen's d	Standardized mean difference	Two-group / t-test comparisons
Hedges' g	Bias-corrected standardized mean difference	Small samples; meta-analysis
Glass's Δ	Mean difference scaled by control-group SD	When group variances differ
Eta-squared (η^2)	Proportion of variance explained by a factor	ANOVA
Partial eta-squared ($\eta^2 p$)	Variance explained, partialling other factors	Factorial ANOVA (common SPSS default)

Measure	What it quantifies	Typical context
Omega-squared (ω^2)	Less-biased variance-explained estimate	ANOVA; preferred over η^2 for inference
Cohen's f / f^2	Effect size for ANOVA / regression	Used as input to power analysis
R^2 / adjusted R^2	Variance explained by a regression model	Adjusted R^2 penalizes extra predictors
Cramér's V / phi (φ)	Association strength in contingency tables	Categorical (also Section 4)
Odds ratio (OR)	Ratio of odds between groups	Logistic regression; case-control studies
Risk ratio / relative risk (RR)	Ratio of probabilities between groups	Cohort studies, trials
Hazard ratio (HR)	Ratio of hazard rates over time	Survival / Cox models (Section 13)
Cliff's delta	Non-parametric dominance effect size	Companion to Mann-Whitney
Rank-biserial correlation	Effect size for rank-based tests	Companion to Wilcoxon / Mann-Whitney
Number needed to treat (NNT)	Patients treated to prevent one event	Clinical interpretability of risk differences

22. Statistical power & sample-size determination

Procedures for planning studies and judging their sensitivity. Power is the probability of detecting a true effect, and these methods relate it to sample size, effect size, and the significance level.

Concept / method	Purpose	Notes
Statistical power ($1-\beta$)	Probability of detecting a true effect	Conventionally targeted at ≥ 0.80
A priori power analysis	Determine sample size before data collection	Needs target effect size, α, and power
Post-hoc (observed) power	Power computed from the observed effect	Widely criticized; adds little beyond the p-value
Sensitivity analysis	Minimum detectable effect for given n, α, power	Useful when sample size is fixed
Minimum detectable effect (MDE)	Smallest effect detectable at target power	Common in A/B testing
Sample-size formulas (e.g., G*Power)	Compute required n for a planned test	Test-specific; G*Power is a common tool
Design effect	Inflate n for clustered/complex samples	Multiplies the required sample size
Simulation-based power	Estimate power by Monte Carlo	For complex models lacking closed forms

23. Confidence-interval & estimation methods

Methods for interval estimation — the estimation counterpart to the hypothesis tests above. They express the uncertainty around a point estimate.

Method	Purpose	Notes
Wald interval	Normal-approximation CI for a parameter	Poor coverage for proportions near 0 or 1
Wilson score interval	CI for a proportion	Much better small-sample coverage than Wald
Clopper-Pearson (exact) interval	Exact CI for a proportion	Conservative; guarantees nominal coverage
Agresti-Coull interval	Adjusted CI for a proportion	Simple, with good coverage
Bootstrap CI (percentile, BCa)	CI by resampling	Few distributional assumptions (Section 18)
Profile-likelihood interval	CI derived from the likelihood function	Good for skewed estimates / GLMs
Delta method	Approximate SE/CI for a function of estimates	First-order Taylor expansion
Fieller's theorem	CI for a ratio of two means	E.g., relative potency, ratio estimands

24. ROC / AUC & diagnostic-accuracy measures

Evaluate and compare classifiers and diagnostic or risk-prediction models, separate from a model's significance.

Measure / test	Purpose	Notes
ROC curve	Trade-off of sensitivity vs. 1−specificity	Threshold-independent classifier evaluation
Area under the curve (AUC / c-statistic)	Overall discrimination of a classifier	0.5 = chance, 1.0 = perfect separation
DeLong's test	Compare two correlated AUCs	Standard test for comparing ROC curves
Sensitivity / specificity	True-positive / true-negative rates	Basis of most diagnostic metrics
Youden's J statistic	Choose an optimal classification threshold	Sensitivity + specificity − 1
Precision-recall curve / F1	Performance under class imbalance	Preferred to ROC when positives are rare
Net reclassification improvement (NRI)	Added value of a new predictor	Used in risk-prediction modeling
Brier score	Accuracy of probabilistic predictions	Mean squared forecast error; calibration
Positive / negative predictive value (PPV / NPV)	Probability a result is correct, given the result	Depend on prevalence, unlike sensitivity/specificity
Likelihood ratios (LR+ / LR-)	How much a result shifts disease odds	Combine sensitivity and specificity; prevalence-independent
Pre-test / post-test probability	Update disease probability with a test result	Bayes' rule applied via likelihood ratios

25. Missing-data methods & tests

Detect and handle missing values. The validity of each approach depends on the missingness mechanism (MCAR, MAR, or MNAR).

Method / test	Purpose	Notes
Little's MCAR test	Test whether data are missing completely at random	Chi-squared based
Complete-case (listwise) deletion	Analyze only fully observed cases	Unbiased only under MCAR; loses power
Multiple imputation (MI / MICE)	Impute several datasets and pool results	Valid under MAR; reflects imputation uncertainty
Expectation-maximization (EM)	Maximum-likelihood estimation with missing data	Iterative; can yield single imputations
Full-information maximum likelihood (FIML)	Use all available data directly in estimation	Common in SEM under MAR
Inverse-probability weighting (IPW)	Reweight observed cases by response probability	Handles MAR; pairs with causal methods

26. Causal-inference methods

Estimate cause-and-effect relationships, especially from observational data where confounding threatens naive comparisons. These are study designs and estimators rather than single tests.

Method	Purpose	Notes
Randomized controlled trial (RCT)	Estimate causal effects via randomization	Gold standard; balances confounders in expectation
Propensity-score matching	Balance covariates in observational data	Matches treated/control on estimated propensity
Inverse-probability-of-treatment weighting	Reweight sample to mimic randomization	Sensitive to extreme weights
Instrumental variables (IV / 2SLS)	Estimate effects despite unmeasured confounding	Requires a valid, relevant instrument
Difference-in-differences (DiD)	Effect from pre/post change across groups	Assumes parallel trends absent treatment
Regression discontinuity (RDD)	Effect at a treatment-assignment cutoff	Local estimate near the threshold
Directed acyclic graphs (DAGs)	Encode assumptions; identify what to adjust for	Guides confounder selection
Mendelian randomization	Use genetic variants as instruments	Common in epidemiology
Bradford Hill viewpoints	Guide judging whether association is causal	Strength, consistency, temporality, dose-response, etc.

27. Mediation & moderation analysis

Decompose how and when an effect operates — mediation asks through what mechanism, moderation asks under what conditions.

Method / test	Purpose	Notes
Baron-Kenny causal-steps approach	Establish mediation via a sequence of regressions	Largely superseded by direct indirect-effect tests
Sobel test	Test the indirect (mediated) effect	Assumes a normal sampling distribution; low power
Bootstrapped indirect effect	CI for the indirect effect by resampling	Preferred approach (e.g., PROCESS macro)
Moderation (interaction terms)	Whether an effect varies by a third variable	Test the product term in the model
Moderated mediation	Indirect effects that depend on a moderator	Conditional indirect effects
Structural equation modeling	Estimate mediation among latent variables	Also Section 15

28. Sequential & adaptive designs

Testing frameworks that analyze data as they accrue, allowing valid early stopping — central to clinical trials and modern A/B testing, where naive repeated 'peeking' inflates Type I error.

Method	Purpose	Notes
Sequential probability ratio test (SPRT)	Test continuously as data arrive	Wald's method; stops when evidence is conclusive
Group sequential design	Pre-planned interim analyses	Permits early stopping for efficacy or futility
Alpha-spending functions	Allocate Type I error across interim looks	Controls overall α with multiple analyses
O'Brien-Fleming / Pocock boundaries	Stopping boundaries for interim analyses	Differ in how readily they stop early
Bayesian adaptive design	Update the design using accruing data	Common in modern trials and experimentation
Always-valid p-values / e-values	Valid inference under continuous monitoring	Modern solution to 'peeking' in A/B tests

29. Spatial statistics

Tests and models for data with geographic or spatial structure, where observations near each other tend to be related.

Test / method	Purpose	Notes
Moran's I	Global spatial autocorrelation	Whether nearby values are similar overall
Geary's C	Spatial autocorrelation (local contrasts)	More sensitive to local differences than Moran's I
Getis-Ord Gi*	Local hot-spot / cold-spot detection	Identifies clusters of high or low values
Local Moran's I (LISA)	Local indicators of spatial association	Decomposes the global statistic
Join-count statistics	Spatial autocorrelation of categorical maps	For binary / nominal data

Test / method	Purpose	Notes
Ripley's K function	Point-pattern clustering across scales	Distinguishes clustering from dispersion
Variogram / kriging	Model spatial dependence and interpolate	Core geostatistical methods
Spatial lag model	Outcome variable regressed on a weighted average of neighboring outcomes (spatial autoregressive term)	Tests whether nearby outcomes influence each other after accounting for covariates; R: spatialreg
Spatial error model	Residual autocorrelation modelled via a spatially lagged error term	Handles spatial clustering in the error structure rather than the outcome; R: spatialreg
Geographically weighted regression (GWR)	Local regression with spatially varying coefficients estimated at each observation point	Detects spatial non-stationarity; coefficients change across space; R: GWR4, spgwr
Spatial cross-validation	Cross-validation that respects spatial autocorrelation by using spatially blocked folds	Standard k-fold leaks spatial information across folds; R: CAST, blockCV

30. Panel-data & econometric tests

Specialized tests for longitudinal (panel) and econometric data, covering model choice, serial correlation, unit roots, cointegration, and instrument validity.

Test	Purpose	Notes
Hausman test	Fixed- vs. random-effects models	Also Section 11
Breusch-Pagan LM test (random effects)	Random effects vs. pooled OLS	Tests whether panel-level variance is zero
Wooldridge test	Serial correlation in panel data	First-difference based
Levin-Lin-Chu test	Panel unit root (common process)	Assumes a common autoregressive parameter
Im-Pesaran-Shin test	Panel unit root (heterogeneous)	Allows individual-specific dynamics
Pedroni / Kao tests	Panel cointegration	Long-run relationships across panels
Sargan-Hansen J test	Overidentifying restrictions in GMM/IV	Assesses instrument validity
Arellano-Bond test	Autocorrelation in dynamic panel GMM	Checks moment-condition validity

31. Psychometrics & item response theory

Methods for measuring latent traits and evaluating the reliability and fairness of scales and tests.

Method / measure	Purpose	Notes
Cronbach's alpha	Internal-consistency reliability	Also Section 16
McDonald's omega	Reliability based on a factor model	Often preferred over alpha
Kuder-Richardson (KR-20)	Reliability for dichotomous items	A special case of alpha

Method / measure	Purpose	Notes
Rasch model (1PL)	Item difficulty on a latent trait	Assumes equal discrimination across items
2PL / 3PL IRT models	Model item difficulty, discrimination, guessing	Item response theory family
Differential item functioning (DIF)	Whether items behave differently across groups	Test fairness/bias (e.g., Mantel-Haenszel DIF)
Confirmatory factor analysis	Test a hypothesized construct structure	Also Section 15
Mokken scale analysis	Non-parametric IRT scaling	Orders items and persons on a latent trait

32. Estimation frameworks & methods

The underlying principles by which model parameters are estimated. These are not tests but the machinery on which nearly every model above is built.

Method	Purpose	Notes
Maximum likelihood estimation (MLE)	Choose parameters that maximize the likelihood	Foundation of GLMs, survival, SEM, and more
Method of moments	Match sample moments to theoretical ones	Simple; often a starting point for other methods
Generalized method of moments (GMM)	Estimate using moment conditions	Common in econometrics; basis of IV/panel estimators
Ordinary least squares (OLS)	Minimize squared residuals	Best linear unbiased under Gauss-Markov assumptions
Generalized / weighted least squares (GLS/WLS)	Least squares with non-constant/correlated errors	Handles heteroscedasticity or autocorrelation
Restricted maximum likelihood (REML)	Unbiased variance-component estimation	Standard for mixed/hierarchical models
Bayesian estimation (posterior, MAP)	Combine prior and likelihood	Yields full posterior; see Section 18
M-estimation	Generalized estimation minimizing a loss	Framework underlying robust estimators (Section 35)
Shrinkage / empirical Bayes (James-Stein)	Pull estimates toward a common value	Reduces variance; basis of hierarchical shrinkage; justified by Stein's paradox (Section 66)
Kernel density estimation (KDE)	Nonparametric estimate of a density	Smooths data without assuming a distribution

33. Flexible & nonparametric regression

Regression methods that relax the straight-line assumption, letting the data shape the fitted relationship. They bridge the linear models of Section 5 and modern smoothing methods.

Method	Purpose	Notes
Polynomial regression	Capture curvature with power terms	Simple but can behave poorly at the extremes
LOESS / LOWESS	Local weighted smoothing	Flexible nonparametric curve; choose span/bandwidth
Kernel regression (Nadaraya-Watson)	Local averaging weighted by a kernel	Bandwidth controls smoothness
Smoothing / regression splines	Piecewise polynomials joined smoothly	Penalized splines control overfitting
Generalized additive models (GAMs)	Sum of smooth functions of predictors	GLM flexibility with nonlinear terms
MARS	Adaptive piecewise-linear regression	Automatically selects knots/interactions
Quantile regression	Model conditional quantiles	Also Section 5; robust to outliers

34. Model selection & cross-validation

Frameworks for choosing among models and estimating out-of-sample performance, complementing the information criteria in Section 6.

Method	Purpose	Notes
k-fold cross-validation	Estimate out-of-sample error	Splits data into k train/test folds
Leave-one-out CV (LOOCV)	Cross-validation with single held-out cases	Low bias; can be computationally heavy
Nested cross-validation	Unbiased error with tuning inside	Separates model selection from evaluation
Repeated / stratified CV	More stable / class-balanced CV estimates	Reduces variance of the estimate
Mallows's Cp	Penalized fit for linear models	Related to AIC
AIC / BIC / AICc	Information criteria for model choice	Also Section 6; AICc corrects small samples
Bootstrap (.632, optimism)	Resampling-based error estimation	Alternative to cross-validation (Section 18)

35. Robust statistics

Methods designed to remain reliable when outliers or heavy-tailed, contaminated data would distort ordinary estimators.

Method / estimator	Purpose	Notes
Huber M-estimation	Robust regression downweighting outliers	Compromise between OLS and absolute-deviation fits
MM-estimation	High-breakdown, efficient robust regression	Resists a large fraction of outliers
Least trimmed squares (LTS)	Fit ignoring the worst residuals	High breakdown point
Theil-Sen estimator	Robust slope via median of pairwise slopes	Nonparametric; resistant to outliers

Method / estimator	Purpose	Notes
RANSAC	Fit from random consensus subsets	Common in computer vision / heavy contamination
Median absolute deviation (MAD)	Robust measure of spread	Resistant alternative to the standard deviation
Trimmed / Winsorized mean	Robust measure of location	Discards or caps extreme values
Sandwich (Huber-White) standard errors	Variance estimates robust to heteroscedasticity	Cluster-robust variants handle grouped data

36. Epidemiology: measures of disease frequency, association & impact

Core epidemiological quantities. The ratio measures (OR, RR, HR) also appear as effect sizes in Section 21; here they sit alongside frequency and attributable measures and rate adjustment.

Measure	What it quantifies	Notes
Incidence (risk / rate)	New cases over time	Cumulative incidence (risk) vs. incidence rate (per person-time)
Prevalence	Existing cases at a point or period	Point vs. period prevalence
Person-time	Total time at risk across individuals	Denominator for incidence rates
Attack rate	Proportion affected in an at-risk group	Used in outbreaks; secondary attack rate among contacts
Mortality / case-fatality rate	Deaths in a population / among cases	CFR conditions on having the disease
Risk difference / attributable risk	Absolute excess risk from exposure	Complements the ratio measures
Attributable fraction (exposed)	Share of exposed cases due to exposure	(RR - 1) / RR
Population attributable risk / fraction	Disease burden attributable in the population	Depends on exposure prevalence
Number needed to treat / harm	Cases treated to cause one outcome change	Inverse of the risk difference
Direct standardization	Age- (or other) adjusted comparable rates	Applies rates to a standard population
Indirect standardization / SMR	Observed vs. expected events ratio	Standardized mortality/morbidity ratio

37. Epidemiology: study designs

The observational and experimental designs used to estimate exposure-outcome relationships. Randomized trials also appear under causal inference (Section 26).

Design	What it is	Notes
Cohort study	Follow exposed/unexposed groups for outcomes	Prospective or retrospective; yields incidence & RR

Design	What it is	Notes
Case-control study	Compare exposure in cases vs. controls	Efficient for rare outcomes; yields odds ratios
Nested case-control	Case-control sampled within a cohort	Retains cohort efficiency with less measurement cost
Case-cohort study	Cases vs. a random subcohort	Allows estimation of rate ratios
Cross-sectional study	Exposure and outcome measured at once	Yields prevalence; weak on temporality
Ecological study	Group- rather than individual-level data	Risk of the ecological fallacy
Case-crossover	Each case serves as its own control	For transient exposures and acute events
Randomized controlled trial	Experimental allocation of exposure	Gold standard; see Section 26

38. Epidemiology: bias, confounding & validity

The systematic errors that threaten the validity of an estimate. Recognizing and controlling these is the distinctive concern of epidemiological methodology.

Concept	What it is	Practical implication
Selection bias	Distortion from how subjects enter the study	E.g., differential loss to follow-up; Berkson's bias (Berkson's paradox, Section 66)
Information / misclassification bias	Error in measuring exposure or outcome	Non-differential biases toward null; differential either way
Recall bias	Differential accuracy of recalled exposure	Common in case-control studies
Confounding	A third factor distorts the association	Control by design or adjustment (Section 26)
Effect modification / interaction	Effect differs across a third variable's levels	A finding to report, not a bias to remove
Collider bias	Conditioning on a common effect	Can be induced by inappropriate adjustment (DAGs)
Immortal-time bias	Misclassified period during which outcome can't occur	Common in pharmaco-epidemiology
Lead-time / length-time bias	Apparent survival gain from screening	Affects screening-program evaluation
Healthy-worker effect	Workers healthier than the general population	A selection effect in occupational studies

39. Epidemiology: infectious-disease & outbreak metrics

Quantities specific to the dynamics and investigation of communicable disease.

Measure / model	What it quantifies	Notes
Basic reproduction number (R_0)	Average secondary cases from one case (fully susceptible)	$R_0 > 1$ implies potential for an epidemic

Measure / model	What it quantifies	Notes
Effective reproduction number (R_t)	Transmission accounting for immunity/interventions	Tracked in real time during outbreaks
Secondary attack rate	Transmission among susceptible contacts	Measures contagiousness in a setting
Herd-immunity threshold	Immune fraction needed to halt spread	Approximately $1 - 1/R_0$
Serial interval / generation time	Time between successive cases / infections	Inputs to transmission estimates
Incubation period	Exposure to symptom onset	Informs quarantine duration
Case-fatality vs. infection-fatality rate	Deaths among cases vs. among all infected	IFR accounts for undetected infections
Compartmental models (SIR / SEIR)	Model transmission dynamics over time	Susceptible-Infected-Recovered and extensions

40. Descriptive statistics & distribution shape

The summary measures used to describe data before any test is run — location, spread, and shape, including the skewness and kurtosis that characterize departures from normality.

Measure	What it describes	Notes
Mean (arithmetic)	Average of the values	Sensitive to outliers
Geometric / harmonic mean	Averages for growth rates / rates	Geometric for multiplicative data, harmonic for rates
Median	Middle value	Robust to outliers and skew
Mode	Most frequent value	Useful for categorical or multimodal data
Range	Maximum minus minimum	Highly sensitive to extremes
Interquartile range (IQR)	Spread of the middle 50%	Robust; basis of the boxplot
Variance / standard deviation	Average squared / typical deviation	SD is in the original units
Coefficient of variation (CV)	SD relative to the mean	Unitless relative dispersion
Standard error of the mean	Variability of the sample mean	SD divided by $\sqrt{n}$
Quantiles / percentiles	Cut points dividing a distribution	Quartiles, deciles, etc.
Five-number summary	Min, Q1, median, Q3, max	Visualized by a boxplot
Moments (raw, central, standardized)	Quantities summarizing distribution shape	Mean, variance, skewness, kurtosis are the first four
Skewness	Asymmetry of a distribution	0 if symmetric; sign shows tail direction (Pearson, moment, Bowley measures)
Kurtosis (excess)	Tailedness / peakedness	Normal has excess 0; leptokurtic > 0, platykurtic < 0
L-moments	Robust moment-like measures from order statistics	Less outlier-sensitive; common in hydrology / EVT

41. Common probability distributions

The distributions that underlie the tests and models throughout this reference, as sampling distributions, error models, or data-generating processes.

Distribution	Typical use	Notes
Normal (Gaussian)	Continuous, symmetric measurements	Central limit theorem; basis of many tests
Student's t	Means with unknown variance / small n	Heavier tails than the normal
Chi-squared	Squared-normal sums; GOF and variance tests	Right-skewed; one df parameter
F	Ratios of variances; ANOVA and regression	Two df parameters
Bernoulli / binomial	Single trial / count of successes in n trials	Discrete; parameters n, p
Poisson	Counts of rare events per interval	Mean equals variance
Negative binomial	Overdispersed counts	Variance exceeds the mean
Geometric / hypergeometric	Trials to first success / sampling without replacement	Discrete
Uniform	Equal probability across a range	Continuous or discrete
Exponential	Time between Poisson events	Memoryless; constant hazard
Gamma / Weibull	Positive, skewed time-to-event data	Flexible hazard shapes (Sections 13, 44)
Beta	Proportions / probabilities on (0,1)	Conjugate prior for the binomial
Log-normal	Multiplicative positive data	Log of the variable is normal

42. Experimental design (DOE)

Designs that structure data collection to estimate effects efficiently and control nuisance variation — the planning counterpart to the ANOVA and regression analyses earlier.

Design	Purpose	Notes
Completely randomized design	Random assignment of units to treatments	Analyzed by one-way ANOVA
Randomized complete block design	Control a known nuisance factor	Blocking reduces error variance
Latin square design	Control two nuisance factors at once	Rows and columns block simultaneously
Factorial design	Study several factors and their interactions	Full factorial covers all combinations
Fractional factorial design	Screen many factors economically	Higher-order effects become aliased
Split-plot design	Mix hard- and easy-to-randomize factors	Has two error strata
Response surface methodology	Optimize a response over factor settings	Central composite, Box-Behnken designs

Design	Purpose	Notes
Taguchi methods	Robust design against noise factors	Orthogonal arrays; quality engineering
Crossover design	Units receive treatments in sequence	Watch for carryover effects
Repeated-measures / longitudinal	Same units measured over time	Analyzed by RM-ANOVA / mixed models

43. Circular & directional statistics

Methods for angular or cyclic data — compass directions, times of day, phases — where ordinary statistics fail because the scale wraps around.

Test / measure	Purpose	Notes
Circular mean / variance	Summarize angular or cyclic data	Standard location/spread fail on wrapped scales
von Mises distribution	Normal analogue on the circle	Concentration parameter κ
Rayleigh test	Uniformity vs. a single preferred direction	Detects a mean direction
Watson's U^2 test	Goodness-of-fit / uniformity on the circle	More general than Rayleigh
Kuiper's test	Rotation-invariant distributional fit	Circular analogue of Kolmogorov-Smirnov
Rao's spacing test	Uniformity via gaps between observations	Sensitive to multimodal patterns
Watson-Williams test	Compare mean directions across groups	Circular analogue of ANOVA
Mardia-Watson-Wheeler test	Compare circular distributions	Nonparametric, rank-based

44. Extreme-value theory & reliability analysis

Methods for the tails of distributions and for failure/lifetime data — rare floods, maximum loads, component failures — where the average behavior is not the concern.

Method / measure	Purpose	Notes
Generalized extreme value (GEV)	Model block maxima	Gumbel, Fréchet, Weibull as special cases
Block maxima method	Fit extremes from periodic maxima	E.g., annual maximum floods
Peaks-over-threshold (POT)	Model exceedances above a threshold	Uses the generalized Pareto distribution
Generalized Pareto distribution (GPD)	Distribution of threshold exceedances	Core of the POT approach
Return level / return period	Magnitude expected once per T periods	Key output in hydrology / risk
Hill estimator	Estimate the tail index	Quantifies how heavy the tail is
Weibull life-data analysis	Model failure / lifetime data	Shape parameter sets increasing/decreasing hazard
Mean time between failures (MTBF)	Average operating time between failures	Reliability-engineering metric

Method / measure	Purpose	Notes
Bathtub hazard curve	Failure rate across a product's life	Infant mortality, useful life, wear-out phases

45. Machine learning & predictive modeling

Algorithms from the predictive-modeling tradition, included for completeness. The emphasis is prediction rather than inference; evaluate them with the metrics in Section 24 and the cross-validation in Section 34.

Method	Purpose	Notes
Decision tree (CART)	Rule-based prediction via recursive splits	Interpretable; overfits on its own
Random forest	Ensemble of bagged decision trees	Robust, captures interactions; less interpretable
Gradient boosting (XGBoost, LightGBM)	Sequential ensemble of weak learners	Often state-of-the-art on tabular data
Support vector machine (SVM)	Maximum-margin classification / regression	Kernels allow nonlinear boundaries
k-nearest neighbors (k-NN)	Predict from the nearest training cases	Simple; sensitive to scaling and dimensionality
Naive Bayes	Probabilistic classifier assuming independence	Fast; a strong text-classification baseline
Neural networks / deep learning	Flexible nonlinear function approximation	Powerful with large data; many hyperparameters
Ensembles (bagging, boosting, stacking)	Combine models to improve accuracy	Reduce variance and/or bias
Regularization (ridge, lasso, elastic net)	Penalize complexity to curb overfitting	Also Section 5
Gaussian process regression	Nonparametric Bayesian regression	Provides predictive uncertainty
k-means / hierarchical clustering	Unsupervised grouping of cases	Also Section 15

46. Analytical pitfalls & threats to inferential validity

Analysis-stage errors that produce misleading results even when the individual tests are computed correctly. Circular analysis — using the same data to both select and test an effect — is the central case, alongside related non-independence and inference traps.

Pitfall	What goes wrong	Mitigation
Circular analysis (double dipping)	The same data both selects and tests an effect, inflating it	Use independent selection and test data; cross-validate
Data leakage	Test information contaminates model training	Strict train/test separation; fit preprocessing on training only
Overfitting	A model captures noise and fails to generalize	Cross-validation (Section 34); regularization; held-out test set
p-hacking / data dredging	Trying many analyses until one is significant	Pre-registration; correct for multiplicity (Section 10); driven by

Pitfall	What goes wrong	Mitigation
		the Law of Truly Large Numbers (Section 66)
HARKing	Presenting post-hoc hypotheses as if a priori	Pre-register hypotheses; label exploratory analyses
Multiple comparisons / forking paths	Many implicit tests inflate false positives	Control FWER or FDR (Section 10)
Regression to the mean	Extreme measurements drift toward the average on retest	Use control groups; account for baseline
Simpson's paradox	An aggregate trend reverses within subgroups	Examine relevant strata; identify confounders
Survivorship / selection on outcome	Analyzing only 'survivors' distorts inference	Define the full at-risk population
Confounding by indication	Treatment assigned by prognosis distorts the effect	Causal-design methods (Section 26)
Base-rate neglect	Ignoring prevalence when interpreting a test	Use predictive values / Bayes (Section 24)
Ecological / atomistic fallacy	Inferring individuals from groups, or vice versa	Match the level of analysis to the claim
Texas sharpshooter fallacy	Finding patterns in noise after the fact	Specify hypotheses before inspecting the data

47. Functional data analysis (FDA)

Statistical methods for data whose observations are entire functions, curves, or surfaces rather than scalar values — such as time series, spectral profiles, or anatomical tract profiles.

Method / technique	Primary purpose	Notes / key assumptions
Functional principal component analysis (fPCA)	Extract dominant modes of variation in a sample of curves	Generalises PCA to function space; scores are scalar summaries of each curve
Functional linear model (FLM)	Regress a scalar outcome on an entire curve predictor (or vice versa)	Coefficient is itself a function; estimated via penalised splines or fPCA expansion
Functional ANOVA (fANOVA)	Compare mean curves across groups	Non-parametric permutation or L2-norm test statistic; pointwise bands
Functional clustering	Group a sample of curves by similarity	k-means on fPC scores; model-based (mixture of Gaussian processes); hierarchical
Functional regression (concurrent / historical)	Relate two functional variables at the same or lagged time points	Concurrent: $Y(t) \sim X(t)$ pointwise; historical: $Y(t)$ depends on $X(s)$, $s \le t$
Registration / curve alignment	Align curves that vary in phase (timing) before amplitude analysis	Separates amplitude and phase variation; landmark or continuous warping
Functional depth measures	Non-parametric ordering of curves from centre outward	Modified band depth (Lopez-Pintado & Romo); basis for

Method / technique	Primary purpose	Notes / key assumptions
		functional boxplots and outlier detection
Functional data outlier detection	Identify curves that deviate from the bulk of the sample	Functional boxplot; magnitude and shape outliers (Hyndman & Shang)

48. Compositional data analysis (CoDa)

Methods for data constrained to a simplex — vectors of positive parts that sum to a constant (e.g., tissue volume fractions, species proportions, budget allocations) — where standard Euclidean statistics are invalid.

Method / technique	Primary purpose	Notes / key assumptions
Centred log-ratio (CLR) transform	Map a composition to unconstrained Euclidean space by dividing each part by the geometric mean	Removes the sum constraint; resulting vector sums to zero; use for PCA, correlation
Isometric log-ratio (ILR) transform	Orthonormal coordinates on the simplex via sequential binary partitions	Fully Euclidean; preferred for regression and distance-based analyses; choice of partition affects interpretation
Additive log-ratio (ALR) transform	Log-ratios of each part to a reference part	Simple but asymmetric — choice of denominator affects results; not isometric
Aitchison distance	Distance measure between two compositions	Euclidean distance in CLR space; respects simplex geometry; R: compositions
Compositional PCA (CoDa-PCA)	Dimension reduction of multivariate compositional data	PCA applied to CLR-transformed data; biplot shows log-ratio structure
Log-ratio regression	Regress a scalar or compositional outcome on compositional predictors	Apply ILR/CLR before standard regression; coefficients interpreted as log-ratio contrasts
Dirichlet regression	Model a composition as outcome directly	Parametric model on the simplex; R: DirichletReg; alternative to ILR for smaller p
Permutation tests on the simplex (PERMANOVA / ANOSIM)	Group differences in compositional space	Use Aitchison distance as input to permutation-based tests; R: vegan with CLR-transformed data
Compositional missing data (zero replacement)	Handle structural or rounded zeros before log-ratio transforms	Multiplicative zero replacement (Martín-Fernández); Bayesian imputation; essential before any log-ratio

49. Topological data analysis (TDA)

Methods from algebraic topology applied to data — characterising shape, connectivity, and multi-scale structure in point clouds, networks, and function spaces, independent of coordinates or thresholds.

Method / technique	Primary purpose	Notes / key assumptions
Persistent homology	Track the birth and death of topological features (connected components, loops, voids) across a range of scales (filtration)	Output: persistence diagram or barcode; scale-free; robust to noise; Python: Ripser, Gudhi
Vietoris-Rips / Čech filtration	Build a sequence of simplicial complexes from a point cloud by growing balls of increasing radius	Most common filtration for persistent homology on Euclidean data; computationally intensive for large n
Persistence diagram / barcode	Visual summary of topological features across scales	Each point encodes birth and death scale of a feature; points far from diagonal are significant
Wasserstein / bottleneck distance	Distance between two persistence diagrams for statistical comparison	Metric on the space of diagrams; used for hypothesis tests and clustering of topological summaries
Persistent entropy / Betti numbers	Scalar summaries derived from persistence diagrams	Betti_0 = connected components, Betti_1 = loops, Betti_2 = voids; used as features in ML pipelines
Mapper algorithm	Construct a simplified graph (nerve of a cover) summarising high-dimensional data topology	Requires choice of filter function, cover, and clustering; Python: KeplerMapper; useful for exploratory shape analysis
Euler characteristic curve	Alternative topological summary tracking the Euler characteristic across filtration levels	Faster to compute than full persistence; R: TDA; used in neuroimaging connectivity studies
Topological machine learning (persistence images / landscapes)	Convert persistence diagrams into vectorised representations for use in standard ML	Persistence images (Adams et al., 2017); persistence landscapes (Bubenik, 2015); Python: persim, giotto-tda

50. Joint longitudinal–survival models

Models that simultaneously specify a longitudinal biomarker trajectory and a time-to-event outcome linked through shared random effects, correcting for informative dropout and enabling combined inference.

Method / model	Primary purpose	Notes / key assumptions
Shared random-effect joint model	Link a linear mixed-effects model for a longitudinal biomarker to a Cox model for a time-to-event outcome via subject-level random effects	The association parameter α quantifies how the biomarker trajectory affects the hazard; R: JM, JMbayes2
Joint model with current value association	Hazard depends on the current (latent) value of the biomarker trajectory at each event time	Most common association structure; separates true biomarker from measurement error
Joint model with slope association	Hazard depends on the rate of change of the biomarker trajectory	Useful when the trajectory velocity (e.g., rate of cognitive decline) drives the event
Multivariate joint models	Multiple longitudinal outcomes jointly modelled with one or more time-to-event processes	R: JMbayes2; accommodates several biomarkers measured on the same subjects
Bayesian joint models	Full posterior inference for joint longitudinal-survival parameters	Stan or JAGS implementations; enables informative priors on association parameters

Method / model	Primary purpose	Notes / key assumptions
Two-stage (naïve) approach	Fit longitudinal and survival models separately, plugging predicted values from stage 1 into stage 2	Biased because it ignores measurement error and informative dropout; shown here as the incorrect alternative
Landmark analysis	At a landmark time t*, use observed biomarker history up to t* to predict subsequent event risk	Simpler than joint models; avoids model specification but discards pre-landmark event information
Dynamic predictions from joint models	Update individual event-free probability in real time as new longitudinal measurements arrive	Key clinical use case; R: JMbayes2 (dynInfo, survfitJM); essential for personalised monitoring

51. Graphical models & structure learning

Probabilistic graphical models encode conditional independence structure among variables as a graph. Structure-learning algorithms recover that graph from data — a different goal from specifying a DAG a priori (Section 26).

Method / technique	Primary purpose	Notes / key assumptions
Gaussian graphical model (GGM)	Estimate conditional independence among continuous variables via a sparse precision (inverse-covariance) matrix; non-zero entries = edges	R: glasso, huge; Python: sklearn GraphicalLassoCV; sparsity chosen by BIC, StARS, or CV
Graphical Lasso (glasso)	L1-penalised maximum likelihood for sparse precision matrix estimation	Friedman et al. (2008); fast coordinate descent; penalty controls network density
CLIME	Sparse precision estimation via column-wise constrained L1 minimisation	Cai et al. (2011); often faster than glasso for very high p
Ising model / binary MRF	Pairwise Markov random field for binary variables; edges encode conditional dependence	Used in psychopathology networks and binary neuroimaging states; estimated via node-wise logistic regression
PC algorithm (constraint-based)	Learn Markov equivalence class (CPDAG) of DAGs from conditional independence tests	Spirtes et al. (2000); assumes no hidden confounders; R/Python: pcalg, bnlearn
FCI (Fast Causal Inference)	Extend PC to handle hidden common causes; outputs a partial ancestral graph (PAG)	More conservative than PC; o-marks indicate possible latent confounders
GES (Greedy Equivalence Search)	Score-based DAG learning using BIC; alternating forward and backward greedy phases on the MEC space	Chickering (2002); often faster than constraint-based; R: pcalg (ges)
LiNGAM	Identify full DAG (beyond MEC) using non-Gaussianity of residuals via ICA	Shimizu et al. (2006); requires at least one non-Gaussian noise term; R/Python: lingam
NOTEARS	Reformulate combinatorial DAG search as continuous smooth optimisation via acyclicity constraint on matrix exponential	Zheng et al. (2018); Python: notears, gcastle; scalable to moderate p
ERGM (exponential random graph model / p*)	Model distribution over networks as exponential family in graph statistics (edges, triangles, degree)	Estimation via MCMC-MLE; R: ergm; used in social network analysis

Method / technique	Primary purpose	Notes / key assumptions
Stochastic block model (SBM)	Latent community structure; nodes in blocks with block-specific edge probabilities	Spectral, variational EM, and belief-propagation estimators; R: blockmodels; Python: graph-tool
Latent space model	Edge probability as decreasing function of Euclidean distance between latent node positions	Hoff, Raftery & Handcock (2002); R: latentnet; handles transitivity naturally

52. Post-selection & high-dimensional inference

When model selection is data-driven (Lasso, stepwise, cross-validation), standard p-values and CIs are invalid — they ignore the selection event. Post-selection inference restores correct Type I error after adaptive selection.

Method / technique	Primary purpose	Notes / key assumptions
Selective inference (polyhedral lemma)	Condition on the Lasso selection event to derive exact post-selection pivot for selected coefficients	Lee, Sun, Sun & Taylor (2016); R: selectiveInference; computationally exact for fixed-lambda Lasso
Data splitting	Use one half for selection, the other half for inference; valid by independence for any selection rule	Simple and broadly applicable; 50% power loss; valid for any selector including black-box ML
Cross-fitting	K-fold variant: alternate nuisance estimation and inference folds; removes bias from plug-in ML estimators	Core of DoubleML (Section 56); Chernozhukov et al. (2018)
PoSI (post-selection simultaneous inference)	Simultaneous valid CIs for all parameters in all possible submodels that could have been selected	Berk, Brown, Buja, Zhang & Zhao (2013); conservative; R: PoSI
Debiased (de-sparsified) Lasso	Add score-equation correction to Lasso estimator to restore sqrt(n)-normal limiting distribution	Zhang & Zhang (2014); van de Geer et al. (2014); R: hdi; enables valid CIs for individual Lasso coefficients
SLOPE (sorted L1 penalised regression)	Lasso variant with decreasing penalty sequence; controls FDR among regression coefficients simultaneously	Bogdan et al. (2015); R: SLOPE; FDR control in regression analogous to BH in testing
Knockoff filter	Construct synthetic negative-control variables (knockoffs) to control FDR among selected predictors	Barber & Candès (2015); Model-X knockoffs: Candès et al. (2018); R: knockoff; Python: knockpy
Higher Criticism (HC)	Scan ordered p-values for excess small values; optimal detection of sparse mixtures of signals in noise	Donoho & Jin (2004); R: HC; used in GWAS and signal detection
Berk-Jones test	Supremum test on empirical vs null CDF of p-values; more powerful than HC in some sparse regimes	Exact and asymptotic versions; R: BJ; global test alternative to Bonferroni for high-dimensional data
SKAT / SKAT-O	Set-based kernel association test for testing whether any variant in a gene set is associated with an outcome	Wu et al. (2011); handles effect heterogeneity via kernel weighting; R: SKAT; standard in GWAS
Max-T procedure	Multiple testing via the permutation distribution of the maximum t-statistic across all hypotheses	Westfall & Young (1993); strongly controls FWER; R: multtest; computationally intensive

Method / technique	Primary purpose	Notes / key assumptions
e-BH procedure	Benjamini-Hochberg algorithm applied to e-values instead of p-values; controls FDR under arbitrary dependence	Wang & Ramdas (2022); complements Section 55 on e-values

53. Conformal prediction & distribution-free uncertainty

Conformal prediction constructs prediction sets with finite-sample, distribution-free coverage guarantees for any black-box predictive model, requiring only that data be exchangeable.

Method / technique	Primary purpose	Notes / key assumptions
Split conformal prediction	Compute nonconformity scores on held-out calibration set; threshold at (1-alpha) quantile gives valid prediction interval	Papadopoulos et al. (2002); exact 1-alpha marginal coverage; R/Python: MAPIE
Full conformal prediction	Refit model for each candidate test label y; computationally intensive but uses all data for calibration	Exact conditional coverage for linear models; efficient algorithms exist for special cases
Cross-conformal prediction	K-fold variant: calibrate using out-of-fold nonconformity scores	Approximate marginal coverage; R: cfcalib; better efficiency than split conformal
Jackknife+ and CV+	LOO and K-fold prediction intervals with guaranteed marginal 1-alpha coverage	Barber, Candès, Ramdas & Tibshirani (2021); R: conformalInference
Mondrian conformal prediction	Stratified conformal: conditional coverage guarantee within each class or stratum	Vovk (2003); ensures equal coverage across positive and negative classes; Python: crepes
Locally weighted conformal	Reweight nonconformity scores by estimated local difficulty; improves conditional coverage	Tibshirani et al. (2019); adapts interval width to local structure
Weighted conformal prediction	Importance-weight calibration scores to handle covariate shift between training and test distributions	Tibshirani et al. (2019); R: weightedconformalprediction
Conformal risk control (CRC)	Control any monotone risk function (FNR, FPR, RMSE) at a user-specified level with finite-sample guarantee	Bates, Angelopoulos et al. (2021); Python: crc; generalises coverage guarantee to arbitrary losses
Risk-controlling prediction sets (RCPS)	Simultaneous control of multiple monotone risk functions at user-specified levels	Angelopoulos et al. (2021); enables e.g. simultaneous FNR and FPR control
Prediction-powered inference (PPI / PPI++)	Valid hypothesis tests and CIs boosted by large unlabelled dataset with ML predictions; finite-sample valid	Angelopoulos et al. (2023); Python: ppi_py; no distributional assumptions
Venn-Abers predictors	Distribution-free calibrated probabilistic classifiers with guaranteed coverage in the Venn prediction framework	Vovk & Petej (2014); stronger calibration than Platt scaling; R: VennAbers

54. Optimal transport & Wasserstein methods

Optimal transport (OT) defines geometrically meaningful distances between probability distributions by finding the minimum-cost coupling (transport plan). It underpins modern methods for comparing, averaging, and regressing on distributions.

Method / technique	Primary purpose	Notes / key assumptions
Wasserstein-1 / -2 distance (Earth Mover)	Minimum expected cost to transport mass from one distribution to another under a ground metric	W1: L1 cost; W2: L2 cost; W2 = Euclidean distance between Gaussians when both are Gaussian; Python: POT, scipy
Sinkhorn algorithm (entropic OT)	Regularise OT with entropy penalty; solve via Sinkhorn-Knopp iterations; O(n^2) per iteration	Cuturi (2013); much faster than exact OT; Python: POT, ott-jax; used in generative modelling
Sliced Wasserstein distance	Average 1D Wasserstein distance over random 1D projections of the distributions	Closed-form for 1D; embarrassingly parallel; scalable alternative to full OT; Python: POT
Wasserstein barycenter	Frechet mean of a set of distributions under W2 distance; unique under mild conditions	Agueh & Carlier (2011); Python: POT; used for averaging connectivity matrices and distributions
Distributional regression (Wasserstein regression)	Regress a distribution-valued outcome on scalar/vector predictors in Wasserstein space	Petersen & Muller (2019); R: fdadensity; extends classical regression to distribution-valued responses
Maximum Mean Discrepancy (MMD)	Non-parametric two-sample test via RKHS kernel embedding distance between distributions	Gretton et al. (2012); calibrated via permutation; Python: torch-two-sample; used for generative model evaluation
Energy distance / Cramer test	Non-parametric two-sample test via expected inter- vs intra-group pairwise distances	Szekely & Rizzo (2004); R: energy; multivariate extension of the Kolmogorov-Smirnov test
OT for domain adaptation / dataset shift	Align source and target distributions by computing OT map; correct for covariate shift	Courty et al. (2017); Python: POT; used in transfer learning and fairness
Frechet Inception Distance (FID)	W2 distance between Gaussians fit to feature distributions of real vs generated samples	Heusel et al. (2017); standard metric for evaluating generative image models

55. E-values, testing by betting & safe inference

E-values are non-negative random variables with expectation <= 1 under the null, interpretable as wealth when betting against the null. They compose multiplicatively, are valid under optional stopping, and underpin the modern testing-by-betting framework.

Method / technique	Primary purpose	Notes / key assumptions
E-value (definition)	Non-negative statistic E with E_H0[E] <= 1; measures evidence against null as bettor's wealth	Vovk & Wang (2021); Bayes factors are e-values; 1/E is a valid p-value by Markov inequality
E-process / test martingale	Cumulative product of e-values forming a non-negative supermartingale under H0	Ramdas, Grunwald et al. (2023); Ville's inequality: P(sup_t E_t >= 1/alpha) <= alpha
Optional stopping validity	E-process remains valid at any data-dependent stopping time without alpha correction	Fundamental advantage over p-values; enables valid sequential monitoring; R: safestats
Anytime-valid sequential t-test	t-test formulated as an e-value; valid at any sample size without group sequential boundaries	Howard et al. (2021); based on mixture likelihood ratio; Python: anytime-valid

Method / technique	Primary purpose	Notes / key assumptions
GRAPPA / GROW criterion	Optimal e-value: maximise expected log-wealth under the alternative (= minimise KL divergence)	Waudby-Smith & Ramdas (2022); analogue of Neyman-Pearson optimality for e-values
e-BH procedure	Apply Benjamini-Hochberg to e-values; controls FDR under arbitrary dependence	Wang & Ramdas (2022); valid without positive regression dependence assumption
Merging e-values	Combine independent e-values via arithmetic mean (valid) or harmonic mean (heavy-tail robust)	Vovk & Wang (2021); arithmetic mean is valid; geometric mean requires independence
Safe tests (testing by betting)	Design test as a betting strategy that is valid (e-value) and powerful (maximises expected log e-value)	Grunwald et al. (2020); R: safestats; unifies SPRT, Bayes factors, and likelihood ratio tests
Universal inference	Split-sample likelihood ratio is an e-value for composite or non-parametric nulls	Wasserman, Ramdas & Balakrishnan (2020); valid without specifying nuisance parameters

56. Causal machine learning & heterogeneous treatment effects

Causal ML combines semi-parametric efficiency theory with flexible ML estimators to estimate average and heterogeneous treatment effects with valid inference, even when the dimension of controls is large.

Method / technique	Primary purpose	Notes / key assumptions
Double/debiased ML (DML)	Partial-linear model: Y = theta*D + g(X) + e; cross-fit ML estimates g(X), valid sqrt(n) inference on theta	Chernozhukov et al. (2018); R: DoubleML; Python: doubleml; handles high-dimensional controls
TMLE (targeted MLE)	One-step correction targeting the causal parameter achieves semi-parametric efficiency bound	van der Laan & Rubin (2006); doubly robust; R: tmle, tlverse
AIPW / doubly-robust estimator	Augmented IPW combining outcome model and propensity model; consistent if either is correctly specified	Robins & Rotnitzky (1995); R: AIPW; Python: econml, zEpid
Causal forest / GRF	Random forest that splits to maximise heterogeneity in treatment effect rather than prediction accuracy; valid CIs	Wager & Athey (2018); R: grf; honest splitting; handles binary and continuous treatments
R-learner	Regress outcome residuals on treatment residuals via flexible learner; targets CATE directly	Nie & Wager (2021); R: rlearner; efficient; flexible choice of base learner
X-learner	Cross-fit imputation of individual treatment effects; effective with unequal treatment group sizes	Künzel et al. (2019); R/Python: metalearners, econml
DR-learner	Doubly-robust pseudo-outcome regression for CATE; achieves semi-parametric efficiency bound for CATE	Kennedy (2023); R: dml; Python: econml
Synthetic DiD (SDID)	Combine synthetic control weighting with DiD; robust to violations of the parallel trends assumption	Arkhangelsky et al. (2021); R: synthdid; staggered adoption extensions
Staggered DiD (Callaway-Sant'Anna)	DiD estimator for settings where different units are treated at different times; avoids forbidden comparisons	Callaway & Sant'Anna (2021); R: did; Sun & Abraham (2021) variant

Method / technique	Primary purpose	Notes / key assumptions
Instrumental forest	GRF extension for IV settings; estimates heterogeneous LATE using an instrument	R: grf (instrumental_forest); combines IV identification with forest-based heterogeneity

57. Bayesian nonparametrics

Bayesian nonparametric (BNP) models place priors over infinite-dimensional spaces, letting model complexity grow with data. Core building blocks are the Dirichlet process and related random measures.

Method / technique	Primary purpose	Notes / key assumptions
Dirichlet process (DP)	Prior over probability distributions; realisations are discrete a.s.; parameterised by base measure G0 and concentration alpha	Ferguson (1973); stick-breaking (Sethuraman 1994); expected clusters ~ alpha * log(n)
Chinese restaurant process (CRP)	Exchangeable partition induced by DP; sequential: customer joins table k prop to size, or new table with prob alpha/(alpha+n)	Collapsed characterisation of DP; basis for collapsed Gibbs samplers
Dirichlet process mixture (DPM)	Infinite mixture model: theta_i | G ~ G; G ~ DP; number of components inferred from data	Neal (2000) Gibbs sampler; R: DPpackage, PReMiuM; Python: BNPy
Pitman-Yor process (PYP)	Two-parameter generalisation of DP with discount d; gives power-law cluster size distribution	Foundation of hierarchical PY language models; R: bnptools
Hierarchical DP (HDP)	Share mixture components across groups via top-level DP; group-level DPs draw atoms from common base	Teh et al. (2006); HDP-HMM for infinite-state HMMs; R: HDP; Python: BNPy
Indian buffet process (IBP)	Prior over binary feature matrices with unbounded columns; each row has sparse active features	Griffiths & Ghahramani (2005); used in Bayesian non-negative matrix factorisation and dictionary learning
Beta process	Completely random measure on features; de Finetti mixing measure for the IBP	Used as prior in Bayesian factor analysis and sparse coding models
Polya tree	Prior over distributions using a recursive tree of Beta-distributed binary splits	Lavine (1992); exact posterior in some cases; computationally difficult in high dimensions
Dependent DP / spatially varying DP	Extend DP to collections of distributions that covary across covariates (space, time, treatment arm)	MacEachern (1999); used for spatially varying mixture models and longitudinal nonparametric Bayes
GP classification (BNP view)	GP prior over latent function; probit or logistic link; posterior approximated via Laplace or EP	Rasmussen & Williams (2006); Python: GPyTorch, GPflow; R: kernlab

58. Optimal experimental design (OED)

Optimal design theory selects treatment allocations, covariate combinations, and sample sizes to maximise information about target parameters given a budget and model. Classical alphabetic criteria (D, A, G, I) meet Bayesian and adaptive extensions.

Method / technique	Primary purpose	Notes / key assumptions
D-optimal design	Maximise det(X'X); minimises volume of confidence ellipsoid for all parameters jointly	Atkinson et al. (2007); R: AlgDesign, skpr; equivalence theorem: D = G optimality
A-optimal design	Minimise trace of (X'X)^{-1}; minimises average variance of parameter estimates	R: AlgDesign; useful when all parameters are equally important
G-optimal design	Minimise maximum prediction variance over the design space	Equivalent to D-optimality by Kiefer-Wolfowitz theorem; preferred when prediction coverage is the goal
I-optimal (IV-optimal) design	Minimise integrated prediction variance over the region of interest	R: AlgDesign, skpr; preferred for response surface optimisation
Bayesian optimal design (BOED)	Maximise expected information gain (EIG = mutual information between parameters and data) averaged over the prior	Chaloner & Verdinelli (1995); Python: BED; computationally intensive; neural estimators of EIG
SMART design	Sequentially randomise participants multiple times across stages to estimate optimal dynamic treatment regime	Murphy (2005); R: DTR, DynTxRegime; key design for personalised medicine (Chapter 57 of textbook)
Adaptive enrichment design	Enrich the trial sample mid-study to focus on subgroups showing evidence of benefit	FDA-endorsed; Bayesian and frequentist versions; connects to platform trials
Multi-armed bandit allocation	Allocate participants to arms probabilistically based on accruing evidence; Thompson sampling, UCB	Lai & Robbins (1985); R: contextual; Python: vowpalwabbit; balances exploration and exploitation
Platform trial / master protocol	Single infrastructure with multiple treatments entering and leaving; perpetual experimental platform	RECOVERY, REMAP-CAP; master protocol with modular sub-studies; group sequential boundaries per sub-study
Coordinate-exchange algorithm	Algorithmic construction of optimal designs by cycling through design points and optimising each	Meyer & Nachtsheim (1995); R: AlgDesign; handles arbitrary design regions

59. Point processes & event sequence models

Point processes model the random occurrence times of discrete events in continuous time. They generalise Poisson processes to allow clustering, inhibition, and history-dependence.

Method / technique	Primary purpose	Notes / key assumptions
Homogeneous Poisson process	Constant intensity lambda; inter-event times exponential; events independent	Foundation model; rate MLE = n/T; KS test on inter-arrivals; R: spatstat
Inhomogeneous Poisson process	Time-varying intensity lambda(t); events conditionally independent given the intensity	Log-likelihood involves integral of lambda(t); rate estimated via kernel smoothing or splines
Cox process (doubly stochastic Poisson)	Poisson process with random intensity function drawn from a stochastic process (e.g., GP)	Captures overdispersion; Bayesian MCMC; R: spatstat; used in ecology and neuroscience
Hawkes process (self-exciting)	Intensity increases after each event; models contagion, aftershocks, spike trains	Ogata (1978); tractable log-likelihood; R: hawkesbow; Python: tick, PyHawkes

Method / technique	Primary purpose	Notes / key assumptions
Inhibitory / repulsive point process	Intensity decreases after events; models refractoriness and spacing	Negative kernel in Hawkes-type model; used in neuroscience and spatial statistics
Marked point process	Events carry associated marks (magnitudes, types); joint model of times and marks	Used in seismology (ETAS model), finance (tick data), genomics (ChIP-seq peak calling)
Spatial K-function / L-function	Test complete spatial randomness; summarise clustering vs inhibition across distance scales	Ripley (1976); R: spatstat; connects to spatial statistics (Section 29) and TDA (Section 49)
Counting process / Andersen-Gill framework	Unify survival analysis and point processes; each subject contributes an event counting process	Andersen & Gill (1982); Cox model = multiplicative intensity model; foundation of multi-state survival models
Renewal process	Events form a sequence of IID inter-arrival times; generalisation of Poisson	Non-exponential inter-arrivals; used in reliability and queueing; R: renewalCount
Neural spike train analysis	Point process GLMs for neuronal firing rate; tuning curves estimated from covariates	Paninski et al. (2007); R: spikesorting; Python: elephant; overdispersion handled via negative binomial

60. High-dimensional & matrix / tensor methods

Statistical inference when the dimension p approaches or exceeds n, and methods for matrix- and tensor-valued data where standard vectorisation destroys structure.

Method / technique	Primary purpose	Notes / key assumptions
Marchenko-Pastur law (RMT)	Eigenvalue distribution of large sample covariance matrix when p/n -> c; separates signal eigenvalues from noise bulk	Tracy-Widom distribution for largest eigenvalue fluctuations; R: RMTstat; threshold for PCA signal extraction
Ledoit-Wolf / Oracle Approximating Shrinkage (OAS)	Shrinkage estimators for sample covariance matrix; reduce estimation error when p is close to n	Ledoit & Wolf (2004); R: corpcor; Python: sklearn LedoitWolf; analytically optimal linear shrinkage
Sparse PCA	PCA with sparsity constraint on loadings; interpretable principal components in high dimensions	Zou, Hastie & Tibshirani (2006); R: sparsepca; Python: sklearn SparsePCA; penalty on loadings
Matrix completion / nuclear norm minimisation	Recover low-rank matrix from subset of observed entries via nuclear norm (trace norm) penalisation	Candès & Recht (2009); R: softImpute; Python: fancyimpute; used in recommender systems and imaging
CP decomposition (CANDECOMP / PARAFAC)	Decompose a 3-way+ tensor as sum of rank-1 outer products; generalises SVD to higher-order arrays	R: rTensor; Python: tensorly; identifiable under mild conditions unlike matrix factorisation
Tucker decomposition	Core tensor multiplied by factor matrices for each mode; more flexible than CP but less identifiable	R: rTensor; Python: tensorly; used in neuroimaging (subject x voxel x time)
Tensor train / matrix product state (MPS)	Decompose high-order tensor as product of low-rank 3-way tensors; scales to very high dimensions	Oseledets (2011); Python: tensorly; used in quantum physics,

Method / technique	Primary purpose	Notes / key assumptions
		compressed sensing, and functional data
Inference for low-rank matrices	Confidence intervals for singular vectors and low-rank approximations; debiased estimators	Cai & Guo (2021); R: RLR; extends CLT to singular value / vector estimation in regression
Non-negative matrix factorisation (NMF) — statistical testing	Bootstrap or permutation tests for the rank of NMF; significance of components	R: NMF; Python: sklearn; complements exploratory NMF with inferential component counting

61. Measurement error models

When predictor variables are measured with error, naive regression estimators are biased and inconsistent (attenuation toward zero for classical error). Measurement error (ME) methods correct for this.

Method / technique	Primary purpose	Notes / key assumptions
Classical ME model	Observe W = X + U where U ~ (0, sigma_u^2) independent of X; OLS slope attenuated by reliability ratio lambda = sigma_X^2 / (sigma_X^2 + sigma_u^2)	Attenuation bias proportional to error variance; reliability ratio < 1; motivates ME correction
SIMEX (simulation extrapolation)	Add artificial measurement error at increasing levels sigma_u^2 * (1+lambda); extrapolate fitted curve back to lambda = -1 (zero error)	Cook & Stefanski (1994); R: simex; works with any estimation procedure; non-parametric extrapolation
Regression calibration	Replace W with E[X | W, Z] (calibrated predictor) in outcome model; corrects attenuation under classical ME	Carroll et al. (2006); R: rccr; requires calibration subsample, validation data, or repeated measures
BCES (Bivariate Correlated Errors and Intrinsic Scatter)	Measurement-error-corrected regression for heteroscedastic errors and intrinsic scatter in both variables	Akritas & Bershady (1996); R: bces; standard in astronomy and environmental science
Deming / total least squares regression	Minimise squared perpendicular distances; assumes known ratio of error variances in X and Y	R: MethComp; generalises OLS (which minimises only Y-direction errors)
Structural SEM approach to ME	Treat true X as latent variable with multiple indicators; estimate via SEM measurement submodel	lavaan, LISREL; corrects ME by modelling it explicitly; connects to Section 15/22
Berkson ME model	Observe W = X + U where X (the true value) is random; U uncorrelated with W	Set-point model: no attenuation in simple linear regression; relevant for dose-response in experimental settings
ME in logistic regression	Attenuation and non-linear bias in logistic regression; no simple closed-form correction	SIMEX and regression calibration both apply; R: mecor; Bayesian joint model alternative
Misclassification in categorical predictors	Binary or polytomous X measured with error; causes bias in regression (not always toward null)	Correction via known or estimated misclassification matrix; R: misclass; validation data needed

62. Network comparison & graph-level inference

Statistical tests and models for comparing entire networks as objects, testing structural properties, and detecting changes between two or more graphs.

Method / technique	Primary purpose	Notes / key assumptions
Network permutation test	Permute network labels (group assignment) and compute observed vs null distribution of any graph statistic	Valid under label exchangeability; flexible choice of test statistic; R: igraph, NetworkX
DeltaCon	Node-affinity-based graph similarity; handles weighted, directed, and bipartite networks scalably	Koutra et al. (2016); Python: deltacon; robust to node correspondence issues
Adjacency spectral embedding (ASE) test	Embed networks in latent space via SVD; test equality of latent positions using multivariate test	Tang et al. (2017); R: iGraphMatch; valid for latent position and degree-corrected SBM
Graph kernel MMD	Two-sample test comparing distributions over graphs using graph kernels (Weisfeiler-Leman, random walk)	Gretton et al. (2012) extended to graph distributions; Python: grakel; captures structural similarity
Latent space network comparison	Test equality of latent positions across two networks using adjacency spectral embedding + Hotelling T^2	Priebe et al. (2019); R: iGraphMatch; principled test under random dot product graph model
Community detection consistency test	Test whether a detected partition corresponds to true block model signal vs noise	Bickel & Sarkar (2016); likelihood ratio test for SBM; distinguishes informative from spurious clustering
Temporal network change detection	Detect structural change points in a sequence of networks via graph-statistic CUSUM	R: dynnet; monitors changes in degree distribution, clustering, or spectral properties over time
Exponential random graph testing	Likelihood ratio test for ERGM parameters; test whether specific network features are above chance	R: ergm (summary, gof); Monte Carlo goodness-of-fit for observed network statistics

63. Differential privacy & statistical inference

Differentially private (DP) mechanisms release statistics with formal privacy guarantees: adding or removing one individual changes the output distribution by at most a factor of e^epsilon. Statistical validity must be preserved despite the added noise.

Method / technique	Primary purpose	Notes / key assumptions
(epsilon, delta)-differential privacy	Mechanism M satisfies (epsilon, delta)-DP if |log P(M(D) in S) - log P(M(D') in S)| <= epsilon for all adjacent D, D' and all S	Dwork et al. (2006); pure DP: delta = 0; approximate DP: delta > 0; epsilon controls privacy-utility trade-off
Laplace mechanism	Add Laplace(0, sensitivity / epsilon) noise to a real-valued query; satisfies pure (epsilon, 0)-DP	Calibrated to L1 global sensitivity; used for means, counts, histograms; R: diffpriv; Python: diffprivlib
Gaussian mechanism	Add N(0, (sensitivity * sqrt(2 ln(1/delta)) / epsilon)^2) noise; satisfies (epsilon, delta)-DP	Better variance than Laplace for small delta and high-dimensional queries; standard for gradient clipping
Randomised response	Each respondent answers truthfully with probability p, randomly otherwise; local DP protocol	Warner (1965) predates DP; local DP: no trusted curator; used in Apple's WWDC privacy tech

Method / technique	Primary purpose	Notes / key assumptions
DP-SGD	Clip per-sample gradients and add Gaussian noise at each step; trains ML models with (epsilon, delta)-DP guarantee	Abadi et al. (2016); Python: Opacus (PyTorch), TF Privacy; privacy accountant tracks epsilon
Private hypothesis testing	Add noise calibrated to test statistic sensitivity; adjust critical values for inflated variance	Couch et al. (2019); R: diffpriv; valid inference under privacy constraint; power decreases with smaller epsilon
DP synthetic data generation	Generate differentially private synthetic dataset preserving marginal or joint distribution statistics	NIST DP Synthetic Data Competition; Python: SDV + DP, smartnoise-sdk; R: synthpop with DP
Privacy-utility trade-off analysis	Compute minimum n for a DP procedure to achieve target power as function of epsilon	R: diffprivlib; information-theoretic lower bounds on estimation error under DP constraint
Local vs. central DP	Local: each individual randomises own data before sending (stronger privacy, more noise); central: trusted curator randomises aggregate	Apple and Google use local DP; central DP requires trust in curator but achieves much higher accuracy

64. Simulation-based & likelihood-free inference

When the likelihood is intractable but data can be simulated from a model, neural network density estimators approximate the posterior, likelihood, or likelihood ratio directly from simulator output.

Method / technique	Primary purpose	Notes / key assumptions
ABC (Approximate Bayesian Computation)	Accept parameter draws if distance between simulated and observed summary statistics is below threshold epsilon	Tavare et al. (1997); R: abc, EasyABC; Python: pyabc; choice of summaries is critical
SNPE (Sequential Neural Posterior Estimation)	Train a normalising flow to approximate the posterior p(theta | x) using simulated pairs; active learning rounds reduce simulator calls	Cranmer et al. (2020); Python: sbi; most sample-efficient SBI method for complex posteriors
NLE (Neural Likelihood Estimation)	Train a conditional density estimator for p(x | theta); use with MCMC for posterior inference	Papamakarios et al. (2019); Python: sbi; modular: swap MCMC sampler freely
NRE (Neural Ratio Estimation)	Train a binary classifier distinguishing (x, theta) from the joint vs marginal; ratio = likelihood ratio	Hermans et al. (2020); Python: sbi; scalable to high-dimensional parameters
Normalising flows	Invertible neural networks mapping simple base distribution to complex target; tractable density and sampling	RealNVP, MAF, NSF, FFJORD; Python: nflows, zuko; used as posterior approximators in SBI
Variational inference (VI)	Approximate posterior with tractable family (e.g., mean-field Gaussian); minimise KL divergence via gradient ascent on ELBO	Jordan et al. (1999); Python: Pyro, NumPyro, Pymc; fast but can underestimate uncertainty
Expectation propagation (EP)	Approximate intractable factors in posterior product via iterative local Gaussian approximations	Minka (2001); Python: GPy; competitive with MCMC for GP classification; converges to global VI in some cases
Score matching / energy-based models	Fit unnormalised model p(x) ~ exp(-E(x)) by matching score function grad log p; avoids partition function	Hyvarinen (2005); denoising score matching (Vincent 2011);

Method / technique	Primary purpose	Notes / key assumptions
		foundation of diffusion generative models
Denoising diffusion probabilistic models (DDPM)	Generative model defined by a Gaussian forward noising process and learned reverse denoising process	Ho et al. (2020); Song et al. (2021) SDE framework; statistical view: reverse SDE = Tweedie posterior mean
Amortised inference	Train a network to map observations to posterior parameters; single forward pass replaces MCMC at test time	VAE encoder as amortised inference; Python: sbi, Pyro; critical for scalable posterior simulation

65. Dynamic treatment regimes & reinforcement learning

Dynamic treatment regimes (DTRs) formalise adaptive sequential decision making in medicine and policy. Reinforcement learning (RL) provides the mathematical framework for learning optimal regimes from data.

Method / technique	Primary purpose	Notes / key assumptions
Dynamic treatment regime (DTR)	Sequence of decision rules d_t(H_t) mapping patient history H_t to treatment A_t; evaluated by expected long-term outcome E[Y^d]	Murphy (2003); optimal DTR d* maximises E[Y^d]; structural nested mean models for semi-parametric DTR inference
Q-learning for DTRs	Backward induction: estimate Q_t(h_t, a_t) at each stage via regression; optimal action = argmax_a Q_t	Murphy (2005); R: DTR, DynTxRegime; standard multi-stage regression approach
A-learning (advantage learning)	Model only the contrast (advantage) between treatments rather than the full Q-function; more efficient	Robins (2004); doubly-robust extensions; R: DynTxRegime
SMART design (statistical perspective)	Sequential multiple assignment randomised trial; randomise at each stage enabling efficient DTR estimation	Murphy (2005); R: DTR; provides data for Q- and A-learning; connects to OED (Section 58)
Off-policy evaluation (OPE)	Estimate value of a target policy using data collected under a different behaviour policy	IPS, doubly-robust OPE, direct method; Python: COBS; R: OfflineRL; fundamental challenge in observational RL
Offline RL (batch RL)	Learn optimal policy from a fixed observational dataset without further experimentation	Levine et al. (2020); conservative Q-learning (CQL) avoids over-optimism; Python: d3rlpy
Markov decision process (MDP)	Sequential decision framework: states S, actions A, transitions P(s'|s,a), rewards R(s,a), discount gamma	Bellman (1957); value iteration, policy iteration; model-based RL requires estimated P; model-free avoids it
Contextual bandit	Single-stage version: choose action, observe immediate reward; balance exploration vs exploitation	Lai & Robbins (1985); Thompson sampling, UCB, epsilon-greedy; R: contextual; Python: vowpalwabbit
Inverse RL / imitation learning	Infer reward function from observed expert demonstrations; learn to replicate expert behaviour	Abbeel & Ng (2004); Python: imitation; useful when reward is unknown but expert data are available
Policy gradient (REINFORCE, PPO)	Directly optimise parameterised policy by gradient ascent on expected return; handles continuous action spaces	Williams (1992); PPO, A3C; Python: stable-baselines3;

Method / technique	Primary purpose	Notes / key assumptions
		connects to stochastic optimisation in statistics

66. Foundational principles & paradoxes of inference

The axioms and formal results that govern how every test and estimator above is interpreted or justified. These are not procedures but the ground rules of statistical evidence itself.

Principle / paradox	Statement	Why it matters
Interpretations of probability	Frequentist: probability as limiting relative frequency in hypothetical repetitions. Bayesian: degree of belief given current information. Propensity: physical disposition of a chance setup	Determines what a p-value, confidence interval, or posterior probability is even allowed to mean; the source of most cross-paradigm confusion
Likelihood Principle	All evidence about θ from the observed data is contained in the likelihood function L(θ)	Birnbaum (1962): implied by sufficiency + conditionality. Bayesian inference obeys it; p-values violate it by averaging over unobserved data
Sufficiency Principle	A sufficient statistic captures all information in the sample about θ; inference should depend on the data only through it	Fisher-Neyman factorization; Rao-Blackwell: conditioning on a sufficient statistic never increases risk. Bedrock of classical test construction
Invariance / equivariance	Inferences should transform coherently under systematic rescaling or reparameterization of data or parameters	Motivates equivariant estimators and invariant tests; MLE is reparameterization-equivariant
Stein's Paradox	For simultaneous estimation of 3+ means under total squared-error loss, the vector of sample means is inadmissible: joint shrinkage always achieves lower total risk	Stein (1956); James & Stein (1961). Formal basis of the shrinkage / empirical-Bayes entry in Section 32 and of hierarchical partial pooling
Berkson's Paradox	Conditioning on a collider (e.g., selection into a clinic or study) induces spurious, typically negative, association between independent traits	Berkson (1946). Mechanism behind hospital-based cohorts finding disease associations absent in the population; see Sections 38 and 26
Law of Truly Large Numbers	With enough trials or a large enough search space, extremely improbable coincidences are virtually certain to occur somewhere by chance	Diaconis & Mosteller (1989). The probabilistic engine behind data dredging, forking paths, and 'miraculous' findings (Section 46)

How to use this reference

Start from the question, not the test. First identify the outcome type (continuous, count, binary, ordinal, time-to-event), the number of groups or predictors, and whether observations are independent or repeated. That combination usually points to one family above. Then check assumptions with the diagnostic tests in Sections 7, 8, and 11, and switch to a robust, non-parametric, or resampling alternative if they are violated. Where several tests apply, report effect sizes and confidence intervals alongside p-values, and pre-specify any multiple-comparison correction.